

A directory of **plots** for genomics in Python

Every chart we'll build in the course, drawn as a single shared visual system — so a volcano plot and a bar chart feel like they belong on the same slide. The last section is ours: bioinformatics ideograms you won't find in a generic chart catalogue.

01 • Amounts

02 • Distributions

03 • Proportions

04 • Relationships

05 • Uncertainty

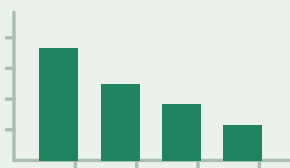
06 • Genomics & bioinformatics

07 • Machine learning

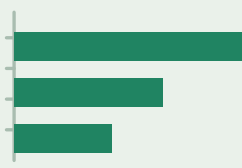
01 Amounts

Numbers for a set of categories — bars, dots, and grids.

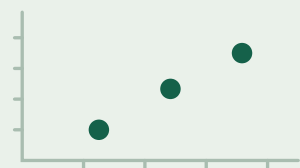
BARS



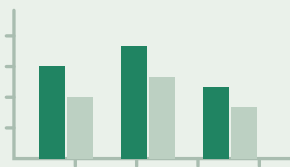
BARS



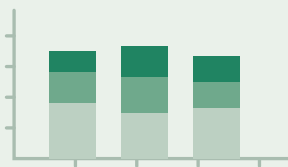
DOTS



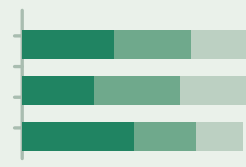
GROUPED BARS



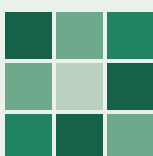
STACKED BARS

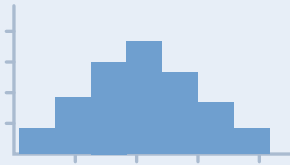
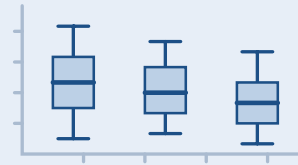
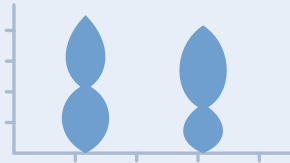
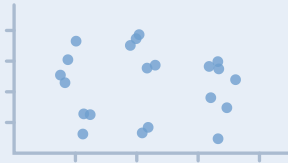
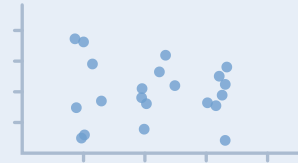
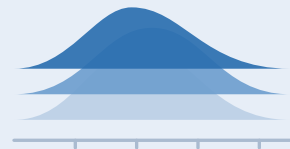
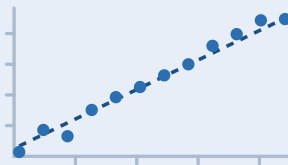
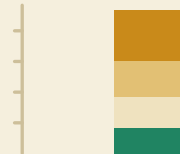
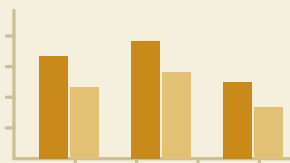


STACKED BARS



HEATMAP



HISTOGRAM**DENSITY****BOXPLOT****VIOLINS****STRIP CHART****SINA****RIDGELINE****Q-Q PLOT****PIE****DONUT****STACKED BAR****GROUPED BARS****MOSAIC****TREEMAP**

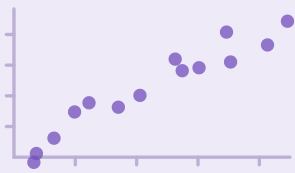
PARALLEL SETS



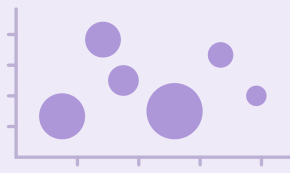
04 x-y relationships

One quantity against another — points, trends, and density.

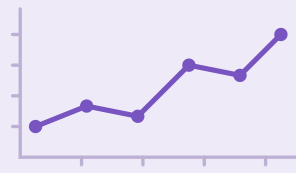
SCATTER



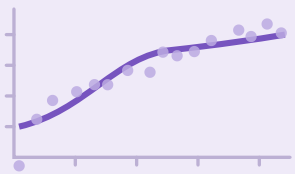
BUBBLE



LINE



SMOOTH (LOESS)



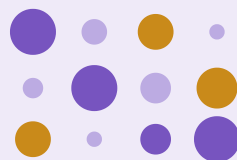
DENSITY CONTOURS



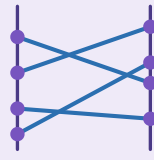
HEX BINS



CORRELOGRAM



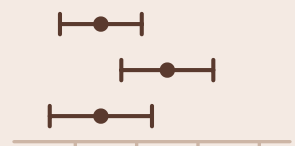
SLOPEGRAPH



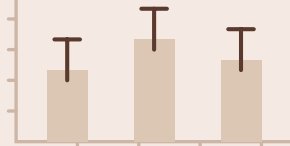
05 Uncertainty

Ranges, intervals, and posterior shapes around an estimate.

ERROR BARS



ERROR BARS



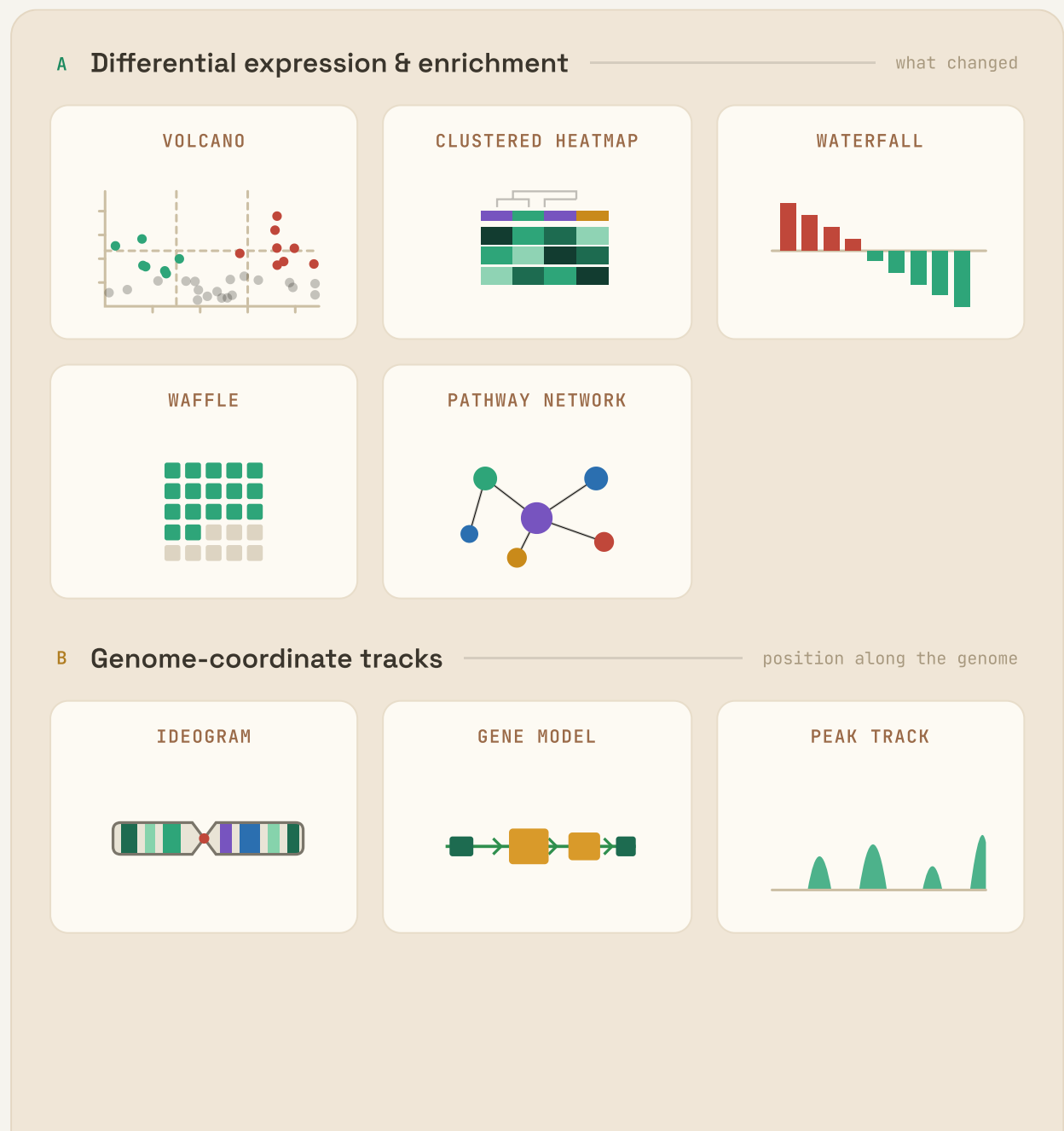
2D ERROR





06 Genomics & bioinformatics

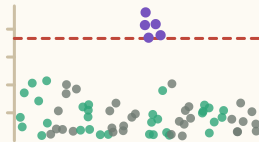
Grouped by what they're for — the question each plot answers.



GENOME COVERAGE



MANHATTAN



C Sequences & alignment

the letters themselves

SEQ. ALIGNMENT



SEQUENCE LOGO



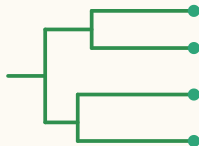
D Comparison, clustering & sets

how groups relate

PCA / UMAP



PHYLOGENY



CIRCOS



SYNTENY



VENN



UPSET



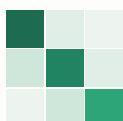
07 Machine learning

Evaluating, training, and explaining models — for the AI-for-graphs day.

A Evaluation

how good is it

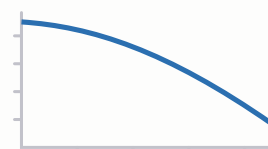
CONFUSION MATRIX

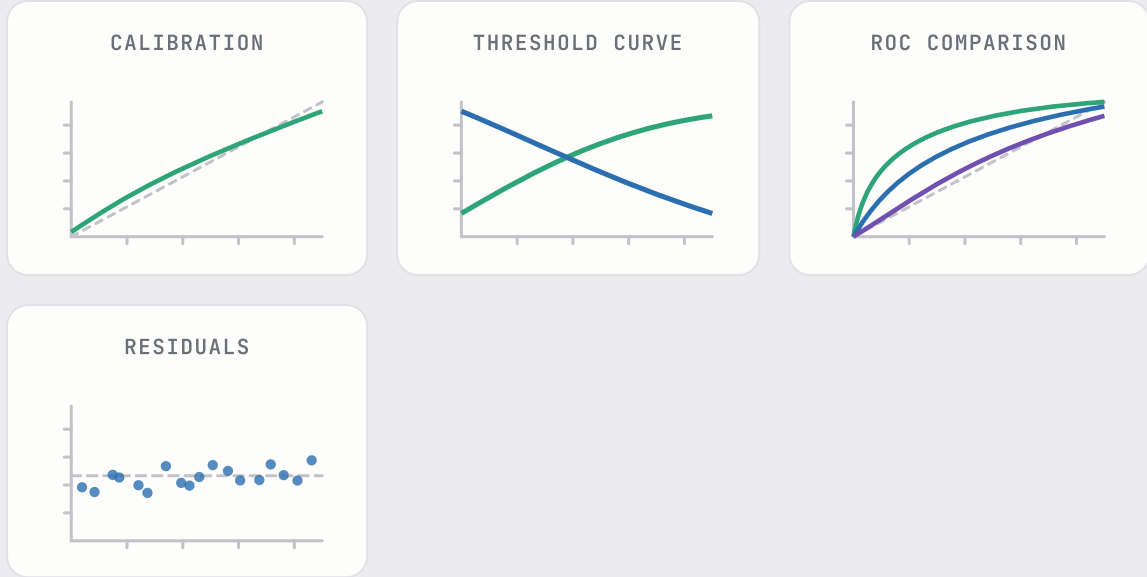


ROC CURVE



PR CURVE





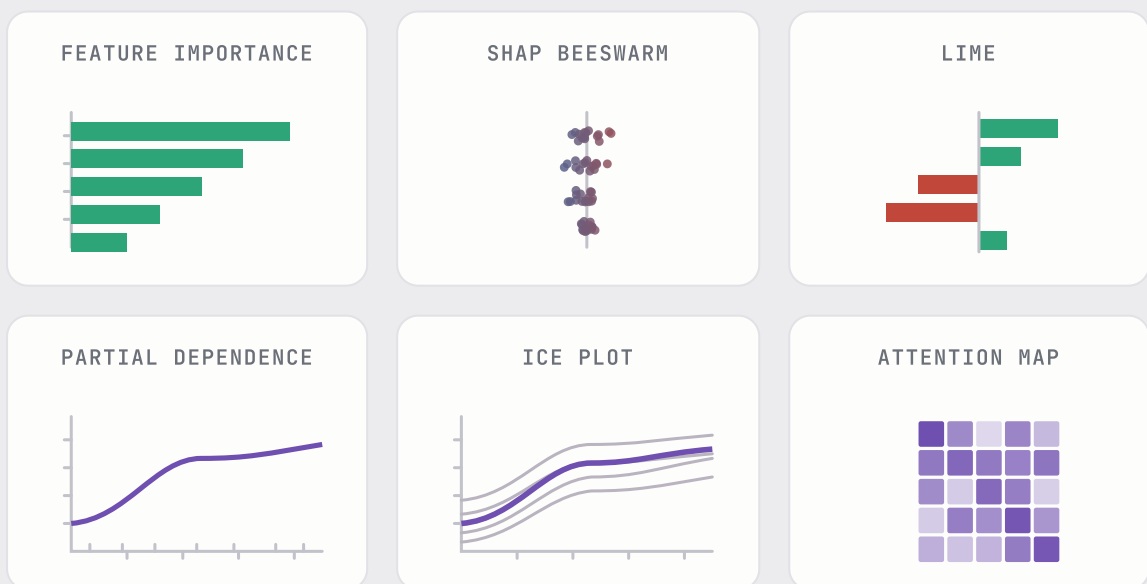
B Training

how it learned



C Interpretation

why it predicts



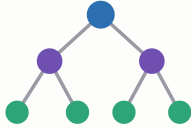
SALIENCY MAP



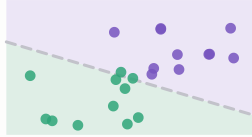
D Models & boundaries

the model itself

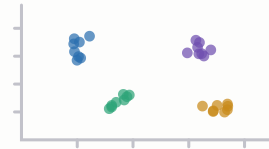
DECISION TREE



DECISION BOUNDARY



EMBEDDING / T-SNE



Genomics Viz with Python • plot directory

One palette • 78 visual idioms • drawn as SVG